

huhusmang / **Awesome-LLMs-for-Vulnerability-Detection** Public

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scripts	fix: fix daily_arxiv.py, the PapersWithCode ap...	11 months ago
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README.md	Add new research papers on software vulner...	2 weeks ago
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Awesome Large Language Models for Vulnerability Detection

Title	Venue	Year	Paper	Github
VulnGym: A Real-World, Project-Level Vulnerability Benchmark for White-Box Vulnerability-Hunting Agents		2026		link
VulTriage: Triple-Path Context Augmentation for LLM-Based Vulnerability Detection		2026	link	link
Synthesizing Multi-Agent Harnesses for Vulnerability Discovery		2026	link	link
QRS: A Rule-Synthesizing Neuro-Symbolic Triad for Autonomous Vulnerability Discovery		2026	link	
SecLens: Benchmarking LLM Vulnerability Detection Through 5 Stakeholder Lenses on 406 Real-World CVEs		2026	link	link
The Semantic Trap: Do Fine-Tuned LLMs Learn Vulnerability Root Cause or Just Functional Pattern?		2026	link	
Sifting the Noise: A Comparative Study of LLM Agents in Vulnerability False Positive Filtering		2026	link	
AgenticSCR: An Autonomous Agentic Secure Code Review for Immature Vulnerabilities Detection		2026	link	
LLM-Based Vulnerability Detection at Project Scale: An Empirical Study		2026	link	
MuVul: Retrieval-Augmented Multi-Agent Code Vulnerability Detection via Cross-Model Prompt Evolution		2026	link	
VulnLLM-R: Specialized Reasoning LLM with Agent Scaffold for Vulnerability Detection		2025	link	
VULPO: Context-Aware Vulnerability Detection via on-Policy LLM Optimization		2025	link	
Specification-Guided Vulnerability Detection with Large Language Models		2025	link	
From Large to Mammoth: A Comparative Evaluation of Large Language Models in Vulnerability Detection	NDSS	2025	link	
Benchmarking LLMs and LLM-based Agents in Practical Vulnerability Detection for Code Repositories	ACL	2025	link	link

Title	Venue	Year	Paper	Github
A Systematic Literature Review on Detecting Software Vulnerabilities with Large Language Models		2025	link	link
LLMxCPG: Context-Aware Vulnerability Detection Through Code Property Graph-Guided Large Language Models	Usenix	2025	link	link
CLeVeR: Multi-modal Contrastive Learning for Vulnerability Code Representation	ACL Findings	2025	link	link
Mono: Is Your "Clean" Vulnerability Dataset Really Solvable? Exposing and Trapping Undecidable Patches and Beyond		2025	link	link
Learning to Focus: Context Extraction for Efficient Code Vulnerability Detection with Language Models		2025	link	
SV-TrustEval-C: Evaluating Structure and Semantic Reasoning in Large Language Models for Source Code Vulnerability Analysis	SP	2025	link	link
SecVulEval: Benchmarking LLMs for Real-World C/C++ Vulnerability Detection		2025	link	link
CVE-Bench: Benchmarking LLM-based Software Engineering Agent's Ability to Repair Real-World CVE Vulnerabilities	NAACL	2025	link	link
R2Vul: Learning to Reason about Software Vulnerabilities with Reinforcement Learning and Structured Reasoning Distillation		2025	link	link
Automated static vulnerability detection via a holistic neuro-symbolic approach		2025	link	
Context-Enhanced Vulnerability Detection Based on Large Language Model		2025	link	
Everything you wanted to know about LLM-based vulnerability detection but were afraid to ask		2025	link	link
MOS: Towards Effective Smart Contract Vulnerability Detection through Mixture-of-Experts Tuning of Large Language Models		2025	link	
Abundant modalities offer more nutrients: multi-modal-based function-level vulnerability detection	TOSEM	2025	link	link
Generative Large Language Model usage in Smart Contract Vulnerability Detection		2025	link	
Closing the Gap: A User Study on the Real-world Usefulness of AI-powered Vulnerability Detection & Repair in the IDE	ICSE	2025	link	link
Vulnerability Detection with Code Language Models: How Far Are We?	ICSE	2025	link	link
Combining Fine-Tuning and LLM-based Agents for Intuitive Smart Contract Auditing with Justifications	ICSE	2025	link	
LAMD: Context-driven Android Malware Detection and Classification with LLMs		2025	link	
LLMs in Software Security: A Survey of Vulnerability Detection Techniques and Insights		2025	link	link
One-for-All Does Not Work! Enhancing Vulnerability Detection by Mixture-of-Experts (MoE)		2025	link	
Leveraging Semantic Relations in Code and Data to Enhance Taint Analysis of Embedded Systems	Usenix	2024	link	link
Effective Vulnerable Function Identification based on CVE Description Empowered by Large Language Models	ASE	2024	link	link
SCALE: Constructing Structured Natural Language Comment Trees for Software Vulnerability Detection	ISSTA	2024	link	link
LLMDFA: Analyzing Dataflow in Code with Large Language Model	NeurIPS	2024	link	link
Learning to Detect and Localize Multilingual Bugs	FSE	2024	link	
GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis	ICSE	2024	link	

Title	Venue	Year	Paper	Github
Sanitizing Large Language Models in Bug Detection with Data-Flow	EMNLP	2024	link	link
RealVul: Can We Detect Vulnerabilities in Web Applications with LLM?	EMNLP	2024	link	
Where is it? Tracing the Vulnerability-relevant Files from Vulnerability Reports	ICSE	2024	link	link
Dataflow Analysis-Inspired Deep Learning for Efficient Vulnerability Detection	ICSE	2024	link	link
Pre-training by Predicting Program Dependencies for Vulnerability Analysis Tasks	ICSE	2024	link	link
Investigating Large Language Models for Code Vulnerability Detection: An Experimental Study		2024	link	link
CleanVul: Automatic Function-Level Vulnerability Detection in Code Commits Using LLM Heuristics		2024	link	link
An Empirical Study of Vulnerability Detection using Federated Learning		2024	link	
LLM-SmartAudit: Advanced Smart Contract Vulnerability Detection		2024	link	link
Advancing Bug Detection in Fastjson2 with Large Language Models Driven Unit Test Generation		2024	link	
Large Language Model for Vulnerability Detection and Repair: Literature Review and the Road Ahead		2024	link	
StagedVulBERT: Multi-Granular Vulnerability Detection with a Novel Pre-trained Code Model		2024	link	link
LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs' Vulnerability Reasoning		2024	link	
Enhancing Source Code Security with LLMs: Demystifying The Challenges and Generating Reliable Repairs		2024	link	
Outside the Comfort Zone: Analysing LLM Capabilities in Software Vulnerability Detection		2024	link	
ANVIL: Anomaly-based Vulnerability Identification without Labelled Training Data		2024	link	
Top Score on the Wrong Exam: On Benchmarking in Machine Learning for Vulnerability Detection		2024	link	
Comparison of Static Application Security Testing Tools and Large Language Models for Repo-level Vulnerability Detection		2024	link	
Vul-RAG: Enhancing LLM-based Vulnerability Detection via Knowledge-level RAG		2024	link	
Security Vulnerability Detection with Multitask Self-Instructed Fine-Tuning of Large Language Models		2024	link	
Generalization-Enhanced Code Vulnerability Detection via Multi-Task Instruction Fine-Tuning	ACL Findings	2024	link	link
M2CVD: Enhancing Vulnerability Semantic through Multi-Model Collaboration for Code Vulnerability Detection		2024	link	link
VulDetectBench: Evaluating the Deep Capability of Vulnerability Detection with Large Language Models		2024	link	link
LLM-Assisted Static Analysis for Detecting Security Vulnerabilities		2024	link	
Multi-role Consensus through LLMs Discussions for Vulnerability Detection		2024	link	link
LLMs Cannot Reliably Identify and Reason About Security Vulnerabilities (Yet?): A Comprehensive Evaluation, Framework, and Benchmarks	IEEE S&P	2024	link	link
Large Language Model for Vulnerability Detection: Emerging Results and Future Directions	ICSE	2024	link	
Prompt-Enhanced Software Vulnerability Detection Using ChatGPT	ICSE	2024	link	
DLAP: A Deep Learning Augmented Large Language Model Prompting Framework for Software Vulnerability Detection		2024	link	

Title	Venue	Year	Paper	Github
Harnessing Large Language Models for Software Vulnerability Detection: A Comprehensive Benchmarking Study		2024	link	
Enhancing Static Analysis for Practical Bug Detection: An LLM-Integrated Approach	OOPSLA	2024	link	link
Source Code Vulnerability Detection: Combining Code Language Models and Code Property Graphs		2024	link	link
Enhancing Code Vulnerability Detection via Vulnerability-Preserving Data Augmentation	LCTES	2024	link	
VulEval: Towards Repository-Level Evaluation of Software Vulnerability Detection		2024	link	
Large Language Model for Vulnerability Detection and Repair: Literature Review and the Road Ahead		2024	link	
A Comprehensive Study of the Capabilities of Large Language Models for Vulnerability Detection		2024	link	
Chain-of-Thought Prompting of Large Language Models for Discovering and Fixing Software Vulnerabilities		2024	link	
Finetuning Large Language Models for Vulnerability Detection		2024	link	link
How Far Have We Gone in Vulnerability Detection Using Large Language Models		2023	link	link
Transformer-based Vulnerability Detection in Code at EditTime: Zero-shot, Few-shot, or Fine-tuning?		2023	link	
Software Vulnerability Detection using Large Language Models	IEEE	2023	link	
DiverseVul: A New Vulnerable Source Code Dataset for Deep Learning Based Vulnerability Detection	RAID	2023	link	
VulBERTa: Simplified Source Code Pre-Training for Vulnerability Detection	IEEE	2022	link	
Deep Learning Based Vulnerability Detection: Are We There Yet?	IEEE	2022	link	
Transformer-Based Language Models for Software Vulnerability Detection	ACSAC	2022	link	
Software Vulnerability Detection Using Deep Neural Networks: A Survey	IEEE	2020	link	
Devign: effective vulnerability identification by learning comprehensive program semantics via graph neural networks	NeurIPS	2019	link	
μVulDeePecker: A Deep Learning-Based System for Multiclass Vulnerability Detection	IEEE	2019	link	
VulDeePecker: A Deep Learning-Based System for Vulnerability Detection	NDSS	2018	link	

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Releases

No releases published

Packages

No packages published

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Languages

